

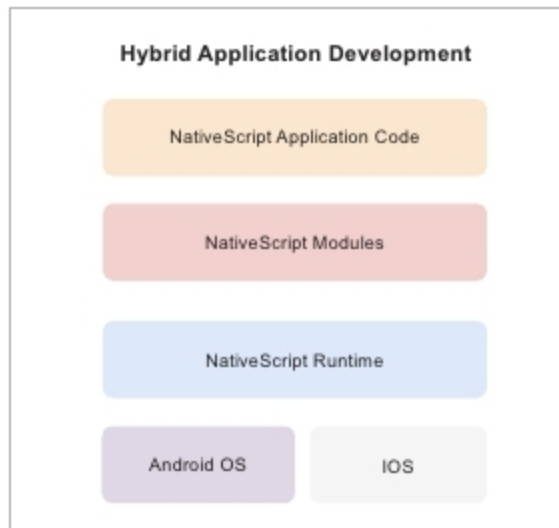


Figure 1: NativeScript architecture

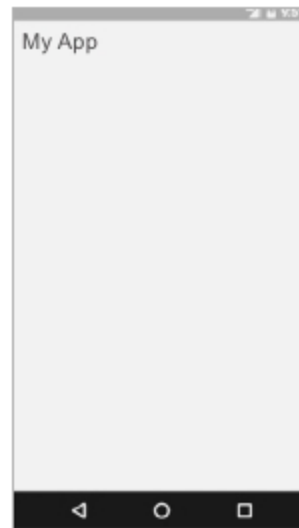


Figure 2: App demo

```
@powershell -NoProfile -ExecutionPolicy Bypass -Command "iex
((new-object net.webclient).DownloadString('https://www.
nativescript.org/setup/win'))"
```

The script will take some time to install everything. Once it is done, you can close the CLI and open it again to refresh it.

After installation, we will have the following:

- Node.js (LTS) 8.x or later
- Google Chrome
- JDK 8
- Android SDK
- Google Repository
- Android SDK build-tools
- Android Studio

The script will work for the Android environment in Windows, and if you want to develop iOS apps in Windows, then NativeScript Sidekick has to be installed. The link for this is <https://www.nativescript.org/nativescript-sidekick>.

Due to time and space constraints, we are skipping this stage.

Setting up the development environment in Mac

systems: If you are using the Mac OS, then you can run the following quick start script:

```
ruby -e "$(curl -fsSL https://www.nativescript.org/setup/
mac)"
```

The script needs administrative access to run many commands, so while executing it, you will often be asked for the password. Once done, all iOS app development prerequisites will be installed.

The next step is to verify the setup. We can use the following command to do so:

```
tns doctor
```

If no issue is detected, it means everything is perfect and we can start our first 'Hello world' kind of application.

Building apps

To create an app, run the following command:

```
tns create HelloOpenSource4u --template nativescript-
template-tutorial
```

Here, the *tns create* command will build an app with the name 'HelloOpenSource4u' and it will use a default template provided by NativeScript named 'nativescript-template-tutorial'. There are other templates also available and you can create your own too. The link for that is <https://github.com/NativeScript/NativeScript/wiki/Using-the-%60tns-create%60-command>.

The *Create* command will take some time, as it fetches all the dependencies and creates a folder with a given app name. Once it is done, go to the folder. In our case, it's 'HelloOpenSource4u'.

Here is the command to run it:

```
cd HelloOpenSource4u
```

Running the app

To start the application, we need to use the *tns run* command inside the folder, as follows:

```
tns run android
```

It will take a few seconds, as NativeScript CLI will build and deploy a native Android application. When the command finishes, the native emulator will open and the app will be seen.

To run the application for iOS, run the command given below:

```
tns run ios
```

Figure 2 shows what happens after the app is run.

The advantages of using NativeScript are:

- It's a hybrid mobile app development platform in which Android and native iOS apps without Web views can be created. It supports cross-platform development.
- It is based on JavaScript, mainly using TypeScript, Angular and Modern JavaScript.
- It offers access to native APIs via JavaScript, so we can even reuse packages from NPM, Gradle and many more.

So give it a try when you develop your next mobile app. **END** 🐧

References

- [1] <https://www.nativescript.org/>
- [2] <https://www.nativescript.org/nativescript-sidekick>

By: Maulik Parekh

The author works at Cisco as a consulting engineer and has an M. Tech in cloud computing from VIT University, Chennai. He constantly strives to learn, grow and innovate. He can be reached at maulikparekh2@gmail.com. Website: <https://www.linkedin.com/in/maulikparekh2>.